

CleanGate Adaptive Verification & Three-Tier Phase Models

VSEPR-SIM Methodology Document
Section: Validation & Thermodynamic Infrastructure

VSEPR-SIM Project

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1 Introduction

This document describes two interconnected subsystems added to the VSEPR-SIM atomistic platform in version 4.0.0:

1. **CleanGate:** A lightweight adaptive verification gate that replaces rigid exhaustive validation with stratified semi-random sampling and risk-driven escalation.
2. **Three-Tier Phase Models:** A selectable hierarchy of melt/gas behavior estimators at different fidelity levels.

Both systems follow the anti-black-box design philosophy: every intermediate score, every sampled module, and every escalation decision is explicitly recorded and traceable.

2 CleanGate: Adaptive Verification

2.1 Motivation

Not every molecular or material case requires the same depth of validation. A simple, stable, well-classified system should pass through quickly. An exotic, thermally stressed, or poorly converging system should receive deeper scrutiny.

CleanGate replaces the “250-check religious ritual” with:

- Stratified semi-random sampling of gate modules across tiers
- Risk scoring via interpretable variables
- Escalation only when justified by measured irregularity

2.2 Gate Tier Registry

Modules are organized into 51 gates across five tiers:

Tier	Range	Example Modules	Character
CORE	0–2	identity, bounds, geometry	Always run
LOW	3–10	formatting, species, charge	Basic sanity
MID	11–25	geometry plausibility, bonding	Chemistry checks
HIGH	26–50	partial relaxation, stability	Physics probes
TARGET	—	case-type specific	Adaptive

Table 1: Gate tier classification with 51 registered modules.

2.3 Semi-Random Stratified Sampling

The sampling strategy balances repeatability with coverage variety:

1. Always include all 3 CORE modules (identity, bounds, geometry)
2. Randomly choose 2 from LOW tier (T3–T10)
3. Randomly choose 2 from MID tier (T11–T25)
4. Randomly choose 2 from HIGH tier (T26–T50)
5. Add 1 targeted module based on detected case type:

- Protein $\rightarrow$ thermal response tendency
- Organometallic $\rightarrow$ coordination number
- Exotic lattice $\rightarrow$ phonon stability
- Thermal stress $\rightarrow$ thermal expansion
- Complex organic $\rightarrow$ torsion range

Total: approximately 10 modules per evaluation (vs. 51 exhaustive).

2.4 Weirdness Score

The weirdness score W measures how far a case departs from the normal envelope of known, stable, classifiable systems:

$$W = w_1 C_{\text{class}} + w_2 G_{\text{geom}} + w_3 E_{\text{energy}} + w_4 B_{\text{bond}} + w_5 R_{\text{ref}} \quad (1)$$

where:

Symbol	Component	Meaning	Default w_i
C	class-uncertain	Hard to classify	0.20
G	geom-irregular	Geometry looks abnormal	0.25
E	energy-conflict	Cheap models disagree	0.20
B	bond-ambiguity	Bonding assignment unstable	0.15
R	reference-distance	Far from known material families	0.20

Table 2: Weirdness score components and default weights (sum = 1.0).

Classification bands:

- 0.00–0.25: ordinary
- 0.25–0.50: unusual
- 0.50–0.75: highly irregular
- 0.75–1.00: exotic or poorly classifiable

2.5 Temperature Score

The temperature score T_s is *not* just temperature—it is thermal severity relative to the material/system:

$$T_s = a_1 \frac{T}{T_{\text{ref}}} + a_2 \frac{|dP/dT|}{S_{\text{expected}}} + a_3 \Phi_{\text{phase}} + a_4 \Phi_{\text{noise}} \quad (2)$$

where:

Symbol	Component	Meaning	Default a_i
T/T_{ref}	temperature ratio	Normalized severity	0.30
$ dP/dT /S$	thermal gradient	Response sensitivity	0.25
Φ_{phase}	phase risk	Transition proximity	0.25
Φ_{noise}	thermal noise	Perturbation instability	0.20

Table 3: Temperature score components and default weights (sum = 1.0).

Classification bands:

- 0.00–0.30: thermally mild

- 0.30–0.60: notable thermal influence
- 0.60–0.80: high thermal stress
- 0.80–1.00: likely phase-risk or decomposition-risk regime

2.6 Convergence Confidence

Estimated from minimization quality signals:

- > 0.80 : strong
- 0.55–0.80: acceptable
- 0.35–0.55: marginal
- < 0.35 : suspicious

Inputs: maximum residual force, RMS force, step count, convergence flag. Excessive step counts (> 5000) incur a penalty indicating a difficult energy landscape.

2.7 Emergence Classification

Three states separate emergence detection from convergence proof:

1. **NONE**: No meaningful emergence detected
2. **VISIBLE_WEAK**: Emergence visible but weakly verified (low convergence)
3. **VISIBLE_STRONG**: Emergence visible and strongly verified

This prevents the classic failure mode: a system looks interesting, everyone gets excited, and six hours later it turns out the solver was hallucinating with confidence.

2.8 Escalation Logic

1. If all prereqs pass and risk is low $\rightarrow$ continue normally
2. If $W > 0.70$ or $T_s > 0.75 \rightarrow$ enable verbose tracking
3. If convergence < 0.45 , or exotic/protein/complex-organic case $\rightarrow$ require deep gate modules
4. If deep gate required and convergence $< 0.30 \rightarrow$ mark result as provisional

2.9 Reporting

Two report formats:

Internal (dense, numeric, machine-readable):

```

1 clean_gate.pass_basic           = true
2 clean_gate.weirdness_score      = 0.82
3 clean_gate.temperature_score    = 0.67
4 clean_gate.convergence_confidence = 0.41
5 clean_gate.prereq_coverage      = 0.58
6 clean_gate.escalate_verbose     = true
7 clean_gate.require_deep_gate    = true

```

External (human-readable summary):

```

1 CleanGate Summary:
2   - Basic prerequisites: PASSED
3   - Material/system irregularity: exotic (0.82)
4   - Thermal severity: high thermal stress (0.67)
5   - Convergence confidence: marginal (0.41)
6   - Deep verification is recommended.

```

3 Three-Tier Phase Models

3.1 Overview

Three selectable modes provide melt and gas behavior estimation at different fidelity/cost trade-offs:

Mode	Name	Melt Method	Gas Method
A	Threshold-Enthalpy	Enthalpy threshold	Ideal + Z -correction
B	Smoothed Phase-Fraction	Sigmoid/piecewise f_ℓ	Cubic EOS (VdW/RK)
C	Stability-Competition	Gibbs G_s vs G_ℓ	Fugacity (RK-based)

Table 4: Phase model modes and their methods.

3.2 Mode A: Threshold-Enthalpy (Fast Screening)

3.2.1 Melting

Accumulated enthalpy:

$$H(T) \approx \int_{T_0}^T c_p(T') dT' \approx c_{p,s} \cdot T \quad (3)$$

Melt onset when $H(T) \geq H_{\text{melt,onset}}$; full melt when $H(T) \geq H_{\text{melt,onset}} + \Delta H_f$.

3.2.2 Gas Behavior

Ideal gas with compressibility correction:

$$PV = ZnRT, \quad Z_{\text{est}} = 1 + \alpha \frac{P}{T} + \beta \rho \quad (4)$$

Best use: very fast prechecks, rough material ranking, large-scale parameter sweeps.

Weakness: melting treated too sharply; poor near phase boundaries.

3.3 Mode B: Smoothed Phase-Fraction (Practical)

3.3.1 Melting

Sigmoid liquid fraction:

$$f_\ell(T) = \frac{1}{1 + \exp[-k(T - T_m)]} \quad (5)$$

or piecewise linear:

$$f_\ell(T) = \begin{cases} 0 & T < T_s \\ \frac{T-T_s}{T_\ell-T_s} & T_s \leq T \leq T_\ell \\ 1 & T > T_\ell \end{cases} \quad (6)$$

Effective heat capacity with latent heat contribution:

$$c_{p,\text{eff}} = (1 - f_\ell) c_{p,s} + f_\ell c_{p,\ell} + \Delta H_f \frac{df_\ell}{dT} \quad (7)$$

Effective density:

$$\rho_{\text{eff}} = (1 - f_\ell) \rho_s + f_\ell \rho_\ell \quad (8)$$

3.3.2 Gas Behavior

Cubic equations of state. Van der Waals:

$$\left(P + \frac{a n^2}{V^2}\right) (V - nb) = nRT \quad (9)$$

Redlich-Kwong:

$$P = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T} V_m (V_m + b)} \quad (10)$$

Newton iteration solvers with deterministic convergence bounds.

Best use: transition-region modeling, materials that soften before full melt, non-ideal gases.

Weakness: still empirical; needs fitted parameters.

3.4 Mode C: Stability-Competition (Serious)

3.4.1 Melting

Gibbs free energy for each phase:

$$G_\phi(T, P) = H_{\phi, \text{ref}} + c_{p, \phi}(T - T_{\text{ref}}) - T \left[S_{\phi, \text{ref}} + c_{p, \phi} \ln \frac{T}{T_{\text{ref}}} \right] + PV_\phi \quad (11)$$

Melt-driving potential:

$$\Delta G_{\text{melt}} = G_\ell - G_s \quad (12)$$

Interpretation: $\Delta G > 0$ means solid favored; $\Delta G < 0$ means liquid favored.

Probabilistic liquid fraction via Boltzmann softening:

$$f_\ell = \frac{1}{1 + \exp(\Delta G_{\text{melt}}/\gamma)} \quad (13)$$

3.4.2 Gas Behavior

Fugacity-based treatment using Redlich-Kwong EOS:

$$\mu = \mu^\circ(T) + RT \ln f, \quad f = \phi P \quad (14)$$

Fugacity coefficient:

$$\ln \phi = (Z - 1) - \ln(Z - B') - \frac{A'}{B'} \ln \left(1 + \frac{B'}{Z} \right) \quad (15)$$

where $A' = aP/(R^2T^{5/2})$, $B' = bP/(RT)$.

Best use: serious phase modeling, exotic materials, publication-grade results.

Weakness: parameter hungry; computationally heavier.

3.5 Output Variables

All three modes produce a unified **PhaseReport** with:

Variable	Range	Meaning
<code>melt_score</code>	$[0, 1]$	Liquid fraction estimate
<code>gas_nonideality_score</code>	$ Z - 1 $	Departure from ideal gas
<code>phase_risk_score</code>	$[0, 1]$	Proximity to phase transition
<code>temperature_severity</code>	$[0, 1]$	Normalized thermal stress

Table 5: Unified output variables across all phase model modes.

4 Integration with CleanGate

The phase model feeds directly into CleanGate’s temperature score:

- `phase_risk_score` $\rightarrow \Phi_{\text{phase}}$ in T_s
- `temperature_severity` provides the T/T_{ref} ratio
- `gas_nonideality_score` contributes to weirdness via energy conflict signals when multiple EOS modes disagree

The `melt_score` also informs CleanGate’s case type detection: systems with $f_\ell \in (0.01, 0.99)$ are automatically tagged as thermal stress cases for targeted module selection.

5 File Manifest

File	Lines	Purpose
<code>atomistic/validation/clean_gate.hpp</code>	~600	CleanGate system
<code>atomistic/core/phase_model.hpp</code>	~540	Three-tier phase models
<code>docs/section_clean_gate_phase_models.tex</code>	this file	Methodology document

Table 6: Files created for CleanGate and phase model infrastructure.

6 Design Principles

1. **Deterministic over theatrical:** All scores are reproducible given the same seed and input data.
2. **Anti-black-box:** Every weirdness component, every sampled module, every Gibbs energy intermediate is exposed and traceable.
3. **Monotonic and interpretable:** Scores increase monotonically with increasing risk/severity. No inversion tricks.
4. **Composable with existing infrastructure:** CleanGate follows the `GateResult/ValidationReport` pattern from `validation_gates.hpp`. Phase models compose with `gas2_eos.hpp` EOS types and `atomistic::thermo` constants.
5. **No “meso” terminology:** All references use “atomistic” per permanent project rule.